README

This document explains the setup procedures and operation methods for the systems developed in this study. I will detail the steps in order so that readers can reproduce the system in their own PC environments. Specifically, I will explain the settings for automatically collecting the YouTube comments to be analyzed, as well as the online environment for managing and executing the programs developed in this study.

Please note that while this document and the programs were created in English as much as possible, some expressions were left in Japanese. This is because forcibly translating Japanese data into English could lead to API rate limits or misinterpretations of the analysis results.Therefore, readers from non-English-speaking regions are encouraged to use tools such as Google Translate.Please also note that the terminology used for various tools may differ slightly depending on when this document is read.

1. Specifying the YouTube channel

In this system, the target YouTube channel for comment data extraction is specified using a "channelId" (a 24-character string starting with "UC" that is uniquely and automatically assigned by the YouTube system). This ID can be found using the following method:

First, open the top page of the target YouTube channel (e.g., UmaMusume Official Channel or Jal Official Sub Channel) in a web browser. Next, right-click on the screen and select "View page source". Then, press Ctrl + F and search for the string "browseId":"UC. The 24-character string starting with "UC" is the unique ID for that channel.

2. Obtaining the YouTube Data API v3 key

2-1. Basic preparation for fetching YouTube comments

Next, obtain the key required for extracting YouTube comment data using the following procedure. First, access the Google Cloud Console (GCC) and log in with your Google account. Next, click "Select a project" at the top of the screen, then click "New project". Enter a Project name of your choice (e.g., youtube-analysis-v2) and click "Create". Make a note of the Project ID that is automatically assigned after creation. Then, from the navigation menu on the top left of the screen, select "APIs & Services" > "Library". Finally, enter "YouTube Data API v3" in the search bar, select the corresponding API, and click "Enable".

2-2. Creating an API Key

From the left navigation menu, open "Credentials" under "APIs & Services". Then, click "+ Create credentials" at the top of the screen and select "API key". Next, in the "Create API key", enter a name of your choice in the "Name", check "Select API restrictions" and choose "YouTube Data API v3" from the dropdown. Under "Application restrictions", choose "None", and then click "Create". Copy the generated API key string("Your API key"). This will be your access key for retrieving comment data from YouTube.

3. Setting Up the BigQuery environment and Issuing a Service account key

Here, you will issue a dedicated key to automatically write data from GitHub to BigQuery in GCP.

3-1. Issuing a service account

From the "Credentials" screen under "APIs & Services" in the GCC left navigation menu, click "+ Create credentials" and select "Service account". Enter a "Service account name" of your choice (e.g., Github-actions) and click "Create and continue". Next, under "Select a role", choose "BigQuery Data Editor" and click "continue". In the "Principals with access (optional)" step, click "Done".

Then, click on the newly created "service Accounts" in the list, navigate to the "Keys" tab, click "Add key", and select "Create new key". Choose "JSON" as the Key type and click "Create". The credentials file will be downloaded to your device. The entire string of text contained within this file will be required for the GitHub configuration described later.

4. GitHub repository configuration and automation process setup

To automate the processes of extracting YouTube data via the API, loading the data into BigQuery, and performing data cleansing, first access GitHub.

4-1. Screens to check on GitLab

On GitHub, access the screen shown in Fig.1, click on “github/workflows” or “main.py”, and verify that the file directory structure matches Fig.2.

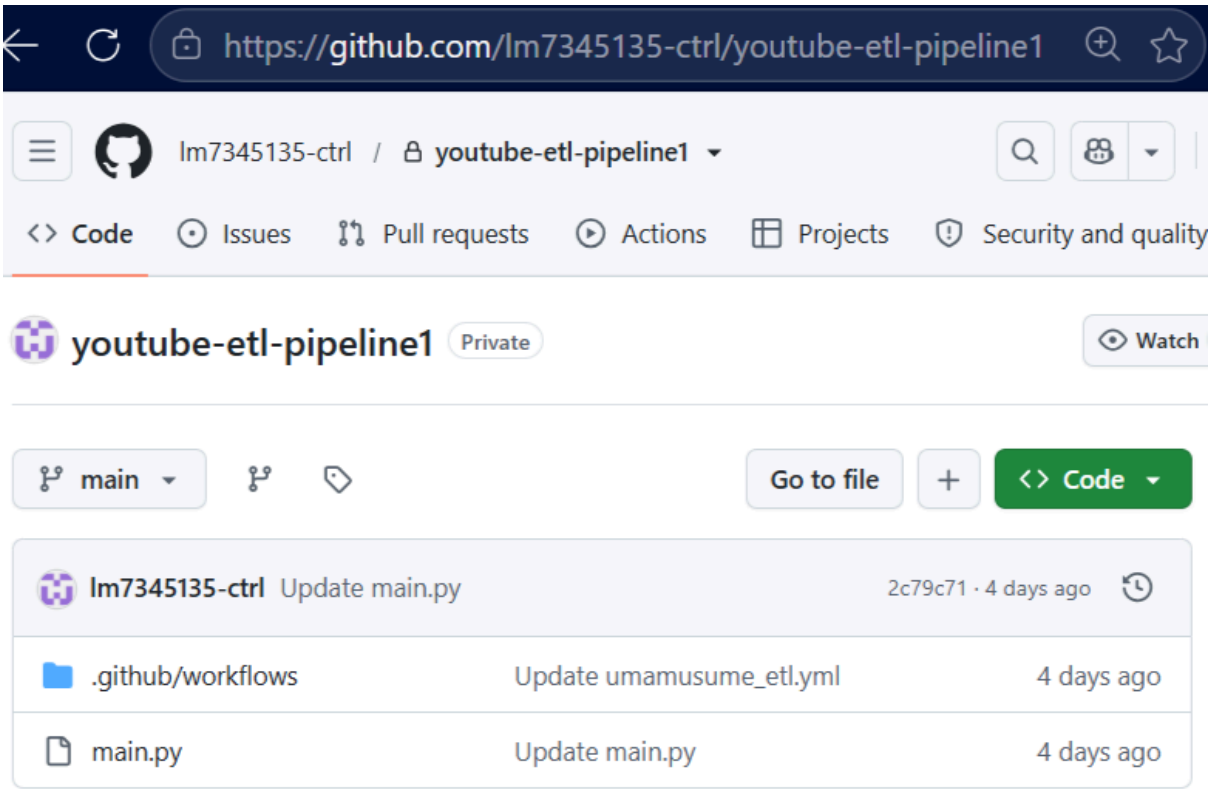


Fig.1 Program management screen(https://github.com/lm7345135-ctrl/youtube-etl-pipeline1)

```
Plaintext

.
├── .github/
│   └── workflows/
│       ├── jal_etl.yml       # Data collection job  definition file for partner companies
│       └── umamusume_etl.yml # Data collection job definition file for target IP
└── main.py                   # Program for API extraction and data transfer to BigQuery
```

Fig.2 Explanation of file

4-2. Description of "main.py"

In "main.py", I implemented the extraction of comment data from the target channel, the data transfer process to BigQuery, and the noise reduction logic for the extracted data. Furthermore, to avoid API rate limits, I introduced a mechanism that progressively extends the wait time before retrying in the event of an error.

4-3. Details of the two yml Files

The automated retrieval schedule for YouTube comments is defined in the cron section (line 4 of the code) of the yml files. To avoid API rate limit errors, the scheduled execution times for each file were intentionally set apart. Note that since the cron schedule in GitHub Actions is based on Coordinated Universal Time (UTC), if you intend to base the schedule on Japan Standard Time (JST), you must specify a time that is 9 hours behind JST. In this system, the schedule for the target IP (UmaMusume) was set to 12:00 UTC (21:00 JST), and for the prospective collaboration company (JAL), it was set to 14:00 UTC (23:00 JST).

Additionally, the transfer destination for the retrieved data is specified within the env section of each file. Specifically, as shown in Fig. 3, the GCP Project ID and the BigQuery dataset name for the GitHub integration are specified here.Note that when reproducing this in your own environment, you should verify the relevant information in GCP or BigQuery and modify it as necessary.

```yaml
YAML

env:
  # Specify the project ID confirmed on the GCP console
  GCP_PROJECT_ID: 'youtube-analysis-v2-499813'

  # Specify the dataset name created in BigQuery
  BQ_DATASET: 'youtube_data'
```

Fig. 3 Relevant information

4-4. Information to register on GitHub

Register the API key and service account key on GitHub using the following steps (see Fig. 4).

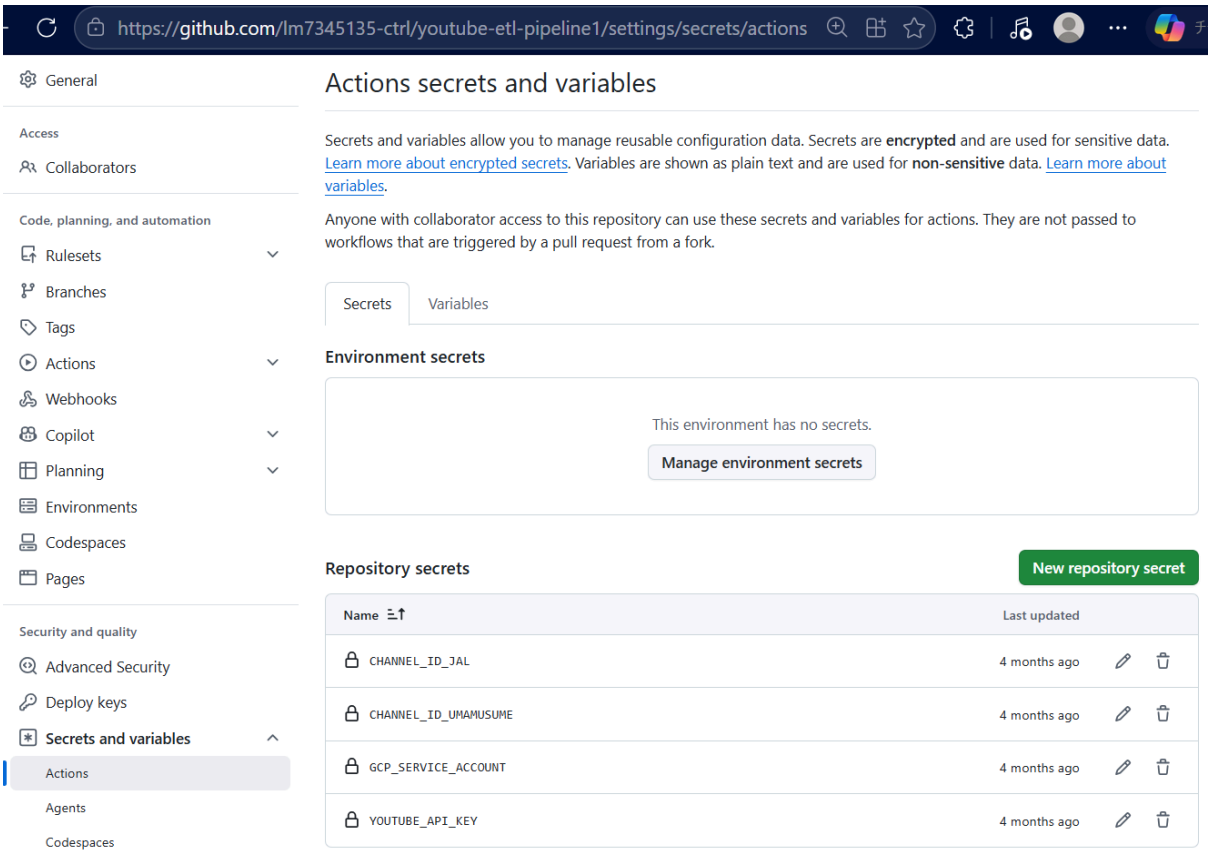


Fig.4 Screen for registering API keys and service account keys in GitHub Secrets. (https://github.com/lm7345135-ctrl/youtube-etl-pipeline1/settings/secrets/actions)

First, open the "Settings" tab in your repository, select "Actions" under "Secrets and variables" from the left menu, and click "New repository secret". Next, enter the variables ① through ④ below into the "Name" and "Secret" fields one by one, clicking "Add secret" each time to complete the registration. By completing this configuration, the entire process—from extracting comment data via the API to structuring it on BigQuery—will be executed autonomously at a specified time every day.

① YouTube API Key

Enter in "Name": YOUTUBE_API_KEY

Enter in "Secret": AIzaSyD_XyZ123...

Explanation: Paste the exact API key string obtained in step 2-2. The "AIza..." string above is a sample.

② BigQuery Service Account Key

Enter in "Name": GCP_SERVICE_ACCOUNT

Enter in "Secret":

```JSON
{
  "type": "service_account",
  "project_id": "youtube-analysis-v2-499813",
  "private_key_id": "abc123def456...",
  "private_key": "-----BEGIN PRIVATE KEY-----\n...\n-----END PRIVATE KEY-----\n",
  "client_email": "github-actions@youtube-analysis-v2-...iam.gserviceaccount.com",
  "client_id": "123456789...",
  "auth_uri": "https://accounts.google.com/o/oauth2/auth",
  "token_uri": "https://oauth2.googleapis.com/token"
}
```

Explanation: Open the file downloaded in step 3-1 with a text editor, copy all the text inside the { }, and paste. The above is sample information.

③ **UmaMusume Official Channel ID**

Enter in "Name": CHANNEL_ID_UMAMUSUME

Enter in "Secret": UCAWxPGGuIfWME2KTLUmOOIQ

Explanation: The 24-character channel ID starting with "UC" identified in step 1. The above is sample information.

④ **Jal Official Sub-channel**

Same as ③. However, use the corresponding Jal information for the "Name" and "Secret".

5. BigQuery data warehouse structure and specifications

This system is designed to separate and manage the comments obtained from YouTube into two distinct areas: an "area for storing raw data" and an "area for managing structured data suitable for analysis." To prevent unexpected billing risks in BigQuery, direct third-party access to the active database is not permitted. As a reference for readers building and replicating this system in their own local environments, the database design specifications will be explained below using screens generated in the author's environment.

5-1. Dataset creation procedure and manual configuration

Log in to the GCC and open the BigQuery console screen from the top-left menu or the search bar (see Fig. 5). Next, click the " ⋮" icon to the right of your project name and select "Create dataset". Finally, enter a desired name in the "Dataset ID" and click "Create dataset".

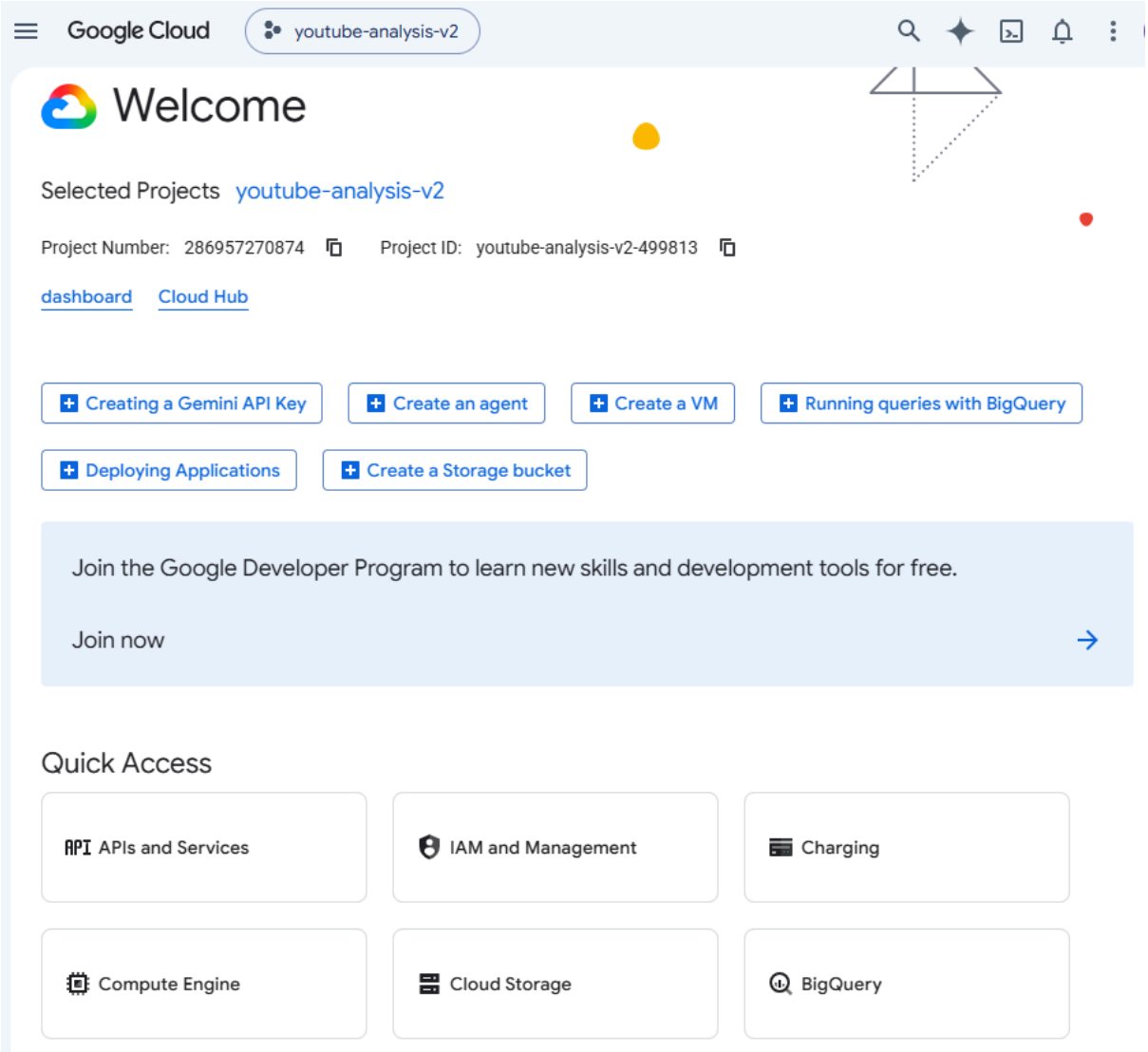


Fig.5 Reference screen for dataset creation

5-2. Storage location for "main.py" execution results

After creating the dataset above, the process from data extraction from YouTube through to data transfer is automatically executed by a script ("main.py"). Upon executing the program, the following three types of aggregated data tables are generated daily.

5-3. raw_comments_brand (raw data table)

This table accumulates all data retrieved from YouTube via the API in its raw, unprocessed form. When "main.py" runs, the latest retrieved comments are appended to the end of the table without deleting or overwriting existing data. Therefore, even if the same comment was retrieved previously, it is stored as an independent record with a different retrieval timestamp (see Table 1).

Table.1 Description of items in the raw data table

Column Name	Data Type	Description
scraped_at	TIMESTAMP	Date and time the data was extracted via API (UTC)
reply_count	INT64	Number of "replies" to the comment
like_count	INT64	Number of "likes" on the comment
brand	STRING	Target brand name for extraction (e.g., Jal or UmaMusume)
channel_id	STRING	YouTube channel ID
published_at	TIMESTAMP	Date and time the comment was posted
video_title	STRING	Video title
text_original	STRING	Original text of the comment
author_channel_id	STRING	Unique ID of the user who posted the comment
video_id	STRING	Unique YouTube video ID
comment_id	STRING	Unique comment ID

5-4. dim_comments_brand (text dimension table)

This table is constructed immediately after the raw data loading is completed, serving to clean noise from raw data as a preprocessing step prior to comment clustering. Specifically, it strips unnecessary HTML tags included in text_original and excludes invalid comments (such as empty strings) that contain no Japanese characters, English letters, or common symbols. Furthermore, if the same comment_id is extracted multiple times, only the chronologically latest record is retained to prevent data duplication (see Table 2).

Table.2 Description of items in the text dimension table

Column Name	Data Type	Description
comment_id	STRING	Unique comment ID
video_id	STRING	Unique YouTube video ID
video_title	STRING	Video title
author_channel_id	STRING	ID of the user who posted the comment
brand	STRING	Target brand name for extraction
text_normalized	STRING	Normalized comment text
text_original	STRING	Original comment text posted by the user
published_at	TIMESTAMP	Date and time the comment was posted
first_scraped_at	TIMESTAMP	Date and time the comment was first extracted by the system

5-5. fct_like_history_brand (engagement fact table)

This table records changes in the number of "likes" and "replies" for each comment over time. Similar to dim_comments, it is constructed immediately after the raw data loading is completed (see Table 3).

Table.3 Description of items in the engagement fact table

Column Name	Data Type	Description
checked_at	TIMESTAMP	Date and time the data was retrieved
comment_id	STRING	Unique comment ID
brand	STRING	Target brand name for extraction (e.g., Jal or UmaMusume)
like_count	INT64	Number of likes at the time of data retrieval
reply_count	INT64	Number of replies at the time of data retrieval

6. Program execution procedures in Google colaboratory

From here, I will explain the procedures for executing programs using Google Colaboratory (Colab) to utilize the massive amount of data accumulated on BigQuery for comment clustering and data analysis for BI tools.

First, access Colab via a web browser and complete the login process. Next, select "Upload notebook". In the displayed window, click "Browse" from the "Upload" tab, select the target program (e.g.,Jal-1004.ipynb and UmaMusume-1004.ipynb), and deploy it to Colab. Please download the target program from the website shown in Fig 6.

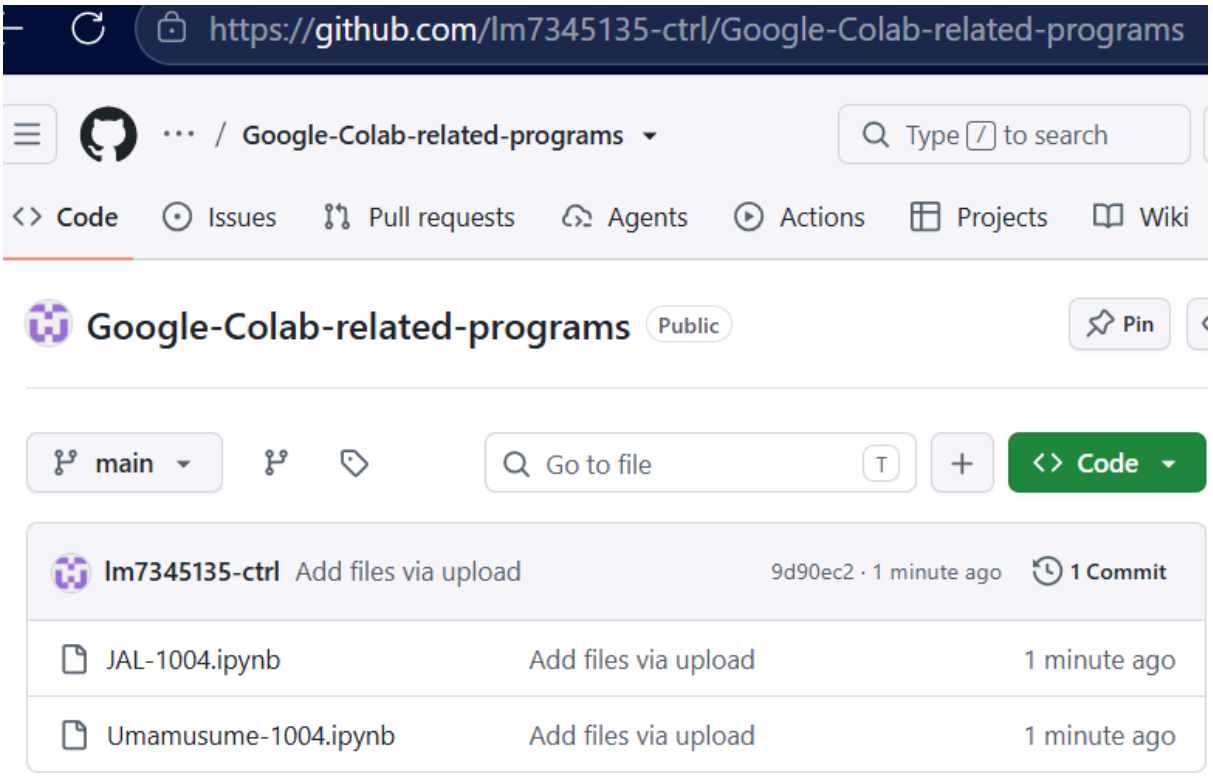


Fig.6 Download the target program from this URL.
https://github.com/lm7345135-ctrl/Google-Colab-related-programs

The program implemented in Colab consists of multiple cells, divided by function. To run the code written in each cell, click the "Run cell" button (right-facing arrow) located at the left end of the cell. For the system to operate properly, it is necessary to press the buttons in order starting from the top cell, confirming the completion of each process before sequentially executing the subsequent cells.

6-1. Cell 1: Data extraction from the data warehouse

In this cell, the data accumulated in BigQuery is deployed to Colab. First, complete the Google account authentication using "auth.authenticate_user()" written in the cell to establish a connection to BigQuery. Then, extract up to 10,000 comment records each from the three types of tables on BigQuery (dim: cleaned data, raw: raw data, fct: numerical history data) in descending order of the comment posting date and time. Upon completion of the process, the number of retrieved comments for each table is output. Furthermore, by displaying the first 5 rows of the dim table in a table format, System visually verifies that there are no missing values or anomalies in the data. Make sure that project_id is the same as the Project ID in Fig.5.

6-2. Cell 2: Comment word segmentation process

When having a computer segment text into words, it is necessary to ensure the system correctly recognizes technical terms and proper nouns while preventing the text from being split into words unintended by the commenter. Therefore, the word segmentation tool "Sudachi" is applied to the comment data retrieved in the first cell. The design specification is set to exclude pronouns and function words, and to detect compound nouns consisting of two or more consecutive nouns. From the detection results, the top 15 words with the highest frequency of appearance in the comments are listed as new word candidates. By selecting arbitrary words from this list and pressing the "Register Selected Words to Dictionary" button, the system is forced to correctly recognize the selected words, preventing them from being improperly segmented. Finally, a word list table (with a file name such as custom_dict.csv) is automatically created in Google Drive so that the processing information is retained even after the system shuts down.

6-3. Cell 3: Deletion of arbitrary words in the word list table

In this function, the dictionary file (custom_dict.csv, etc.) on Google Drive is first loaded, and a list of registered words is displayed. You can then delete specific words by selecting unnecessary terms from the displayed list and clicking the "Delete selected words permanently" button. If there are no words to delete, you can skip this process simply by clicking the "Run cell" button at the left end of the third cell.

6-4. Cell 4: Dynamic clustering using a large language model(LLM)

In this cell, massive amounts of comment data are automatically classified into multiple groups based on the similarity of the comment context. You can also configure the conditions for analyzing the comment dataset. The configuration items consist of the following five stages.

First, at the top of this cell, confirm that the code description for "DATASET_ID" is the dataset ID created in "5-1. Dataset creation procedure and manual configuration" and that the code description for "GCP_PROJECT_ID" is the project ID for "5-1. Dataset creation procedure and manual configuration". If not, fix it. Verify that "0. Select Target Brand" is set to "jal" if the program name is "Jal-1004.ipynb", or "UmaMusume" if the program name is "UmaMusume-1004.ipynb". Next, under "1. Specify Period (Based on comment posting date)", if "PERIOD_MODE" is set to "Recent months (All data)", you can retrieve comment data for the most recent few months. If you select "Specify any period", you can retrieve comment data within the timeframe from "START_DATE" to "END_DATE", so choose your preferred option (refer to Fig. 7).

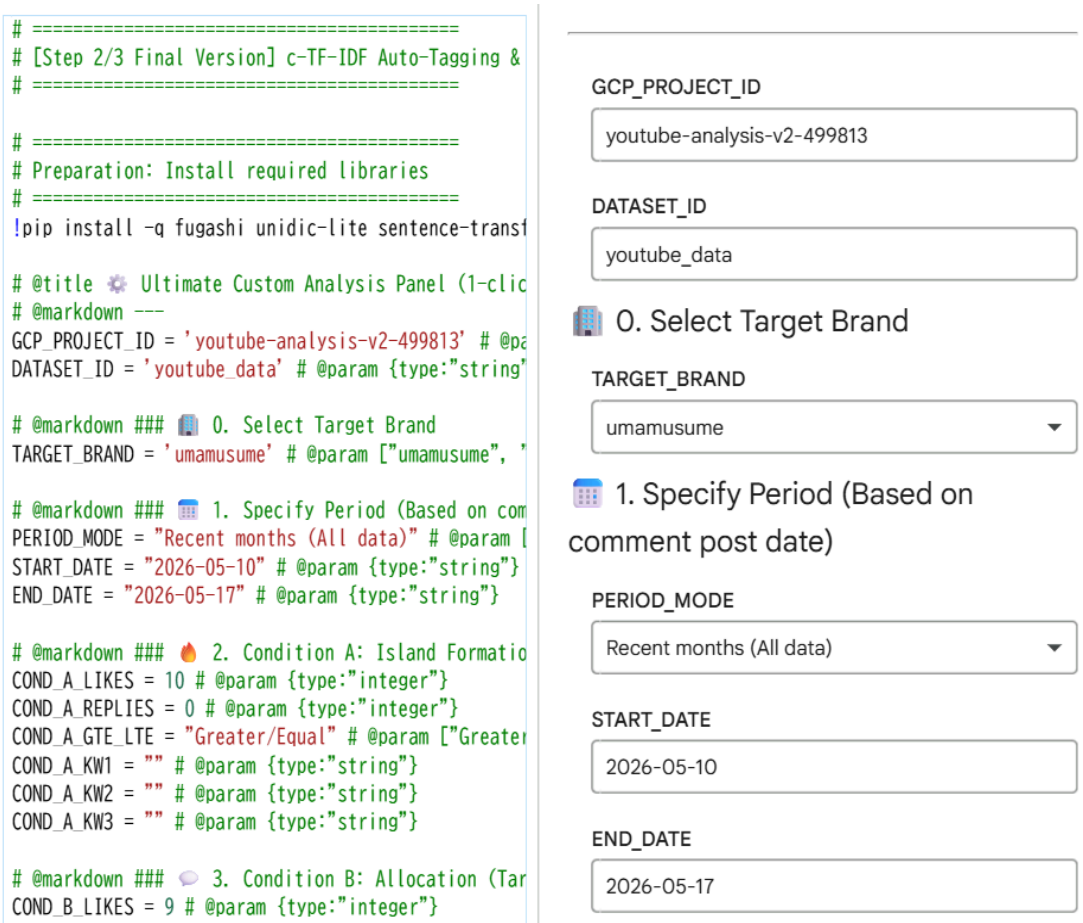


Fig.7 Description of screen items in Cell 4

Next, the details of the clustering method are explained. To avoid out-of-memory errors caused by the massive volume of data processed at one time in Colab, this method introduces a two-stage hierarchical approach. HDBSCAN, a density-based method, is applied as the classification algorithm. HDBSCAN enables the formation of clusters centered around dense areas of comment data arranged on coordinates based on context similarity,

allowing for clear separation between clusters of different topics. Review the following text while referring to Fig. 8.

As the first stage of the two-stage hierarchical approach, comment data satisfying specific conditions (Condition A) is used to form the centers of the clusters.

In the configuration for Condition A, user evaluation metrics—such as the number of "Likes" (COND_A_LIKES) or "Replies" (COND_A_REPLIES) a comment receives—are utilized, and it is possible to specify whether these values fall above or below a certain threshold (COND_A_GTE_LTE).

Furthermore, this can be combined with a condition indicating whether the data contains up to three arbitrary keywords (COND_A_KW1 to 3).

In the second stage, the remaining data is allocated to the existing clusters that were just formed. This allocation logic can be selected from two methods depending on your analysis goals via "ALLOCATION_MODE" within "Island Allocation Logic Selection".

The first method, "1.Overflowed commnts only", allocates to the existing clusters only the residual data that met Condition A but was excluded from the initial clustering due to exceeding the sampling upper limit caused by memory constraints. This is suitable when you want to minimize computational load while focusing your aggregation on highly-rated or influential comments containing specific keywords.

The second method, "2.Overflow + All B comments", collectively distributes both the residual data of Condition A and additional target data satisfying a separately configured Condition B into the existing clusters. This is applied when conducting an exhaustive analysis of the entire comment set. Here, the target Condition B is defined by the parameter group in "Condition B: Allocation (Target Data) Extraction Rules". Similar to Condition A, the comment data allocated to each cluster is controlled by the number of Likes(COND_B_Likes), Replies (COND_B_Replies), their threshold conditions (COND_B_GTE_LTE), and up to three arbitrary keywords (COND_B_KW1 to 3).

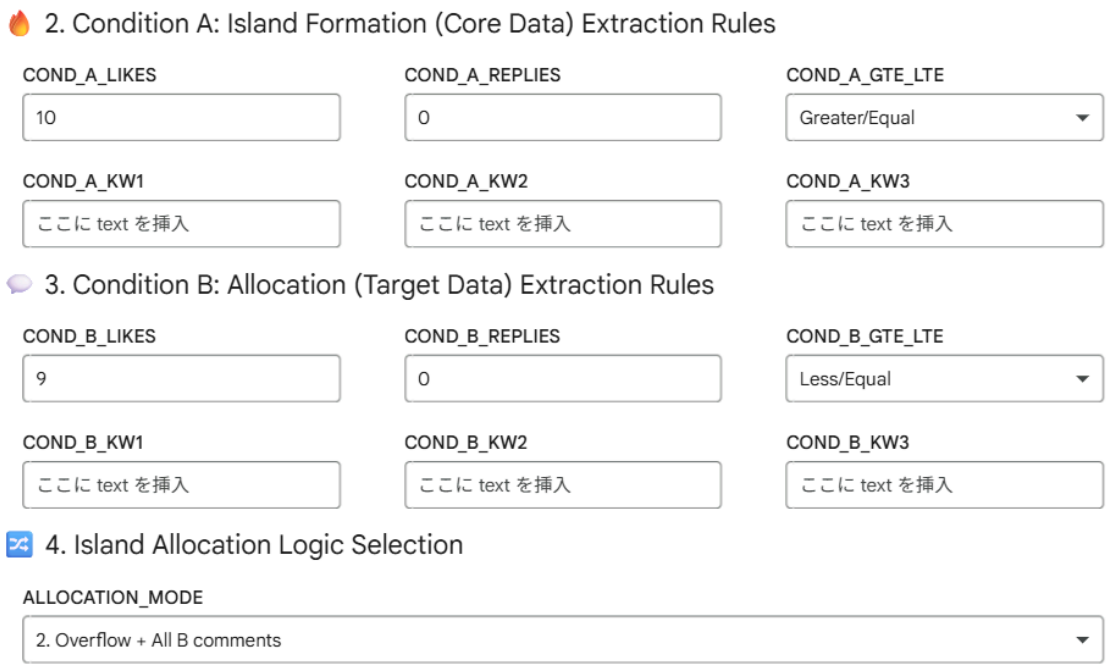


Fig.8 Continued screen description for Cell 4

Subsequently, in "5. Topic Classification Tuning (Adjust with slider)", the granularity of the clustering can be fine-tuned by modifying the following values on the screen (Fig. 9).



Fig.9 Continued screen description for Cell 4

"TOPIC_SCOPE" is an index of the range within which similar comments are integrated into the same cluster. The smaller the value, the more the clusters are subdivided based on minute differences in comment content. "MIN_CLUSTER_SIZE" is an index specifying the minimum number of comment data points required to establish a cluster. For "CLUSTER_PURITY_FILTER", setting this to a lower value results in a cluster composition that prioritizes exhaustive coverage, actively including peripheral comments located around the cluster boundaries.

After configuring the analysis conditions as described above, click the "Run cell" button located at the left end of the cell. Once the analysis is complete, the clustering results for the target comment group are presented. A preview is generated directly below the cell so you can intuitively grasp the output. The definitions of each component are as follows.

"index" is the serial number of each comment data point relative to the entire dataset targeted for analysis. "Cluster_ID" is the identification number of the cluster; comment data groups with the same number share a similar context. Note that noise data failing to meet the set clustering conditions is clearly marked as "-1". "Extracted_Tags" is a group of keywords symbolizing the context of the corresponding "Cluster_ID". For "Purity_Prob", a higher value signifies that the comment is central and symbolic, representing that specific cluster. Within each cluster, the top 3 comments with the highest 'Purity_Prob' scores will be extracted under '[Top 3 Representative Comments]'. Likes and Replies are comment evaluation metrics at the time of data acquisition."Comment_Text" is the actual comment targeted for analysis."Total Processed Comments" is the overall number of comments targeted for analysis."Number of islands formed" is the number of clusters generated. "Valid classified records" is the total number of comments successfully used to form clusters. "Excluded as Noise" is the total number of comments excluded as noise.

6-5. Cell 5: Hybrid search and reinforcement learning function

To efficiently grasp the comment clustering results from the 4th cell, a "keyword search engine" using BM25 and a "context search engine" using FAISS have been implemented.

I will explain how to use these two search engines. First, on the screen shown in Fig. 10, enter any keyword into "Query". Then, when you click "5-Patterns Search", the following items will appear under each of the "Weight 0.0", "Weight 0.25", "Weight 0.5", "Weight 0.75",

and "Weight 1.0" tabs. "RRF_Score" indicates the degree of matching with the search intent. "Cluster_ID" is the cluster number. " Extracted_Tag" is the word symbolizing that cluster. "Likes" is the number of likes for that comment. "Replies" is the number of replies to that comment. "Comment Text" is the content of the comment. Regarding the numerical details for "Weight 0.0", "Weight 0.25", "Weight 0.5", "Weight 0.75", and "Weight 1.0", these represent five patterns where the weighting of the two search engines is varied (the combination ratios of keyword search to context search are 0:1, 0.25:0.75, 0.5:0.5, 0.75:0.25, and 1:0). Next, enter an evaluation score from "0 to 3" for the search results on each tab. When you click "Save Eval & Train AI", the reinforcement learning system using Gymnasium learns this feedback history, and from the next search onwards, the tab with the search ratio best suited to the user's preferences will automatically be placed on the left side. Additionally, you can delete the learning history by clicking "Reset All AI Training Data".

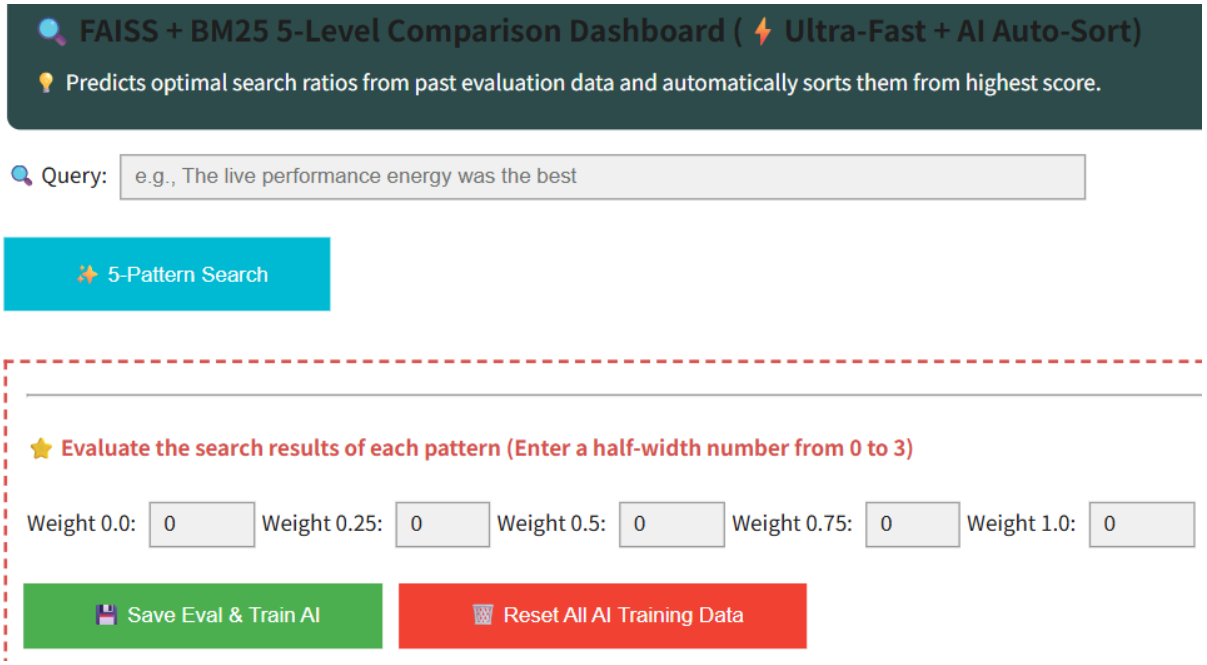


Fig.10 Screen description for Cell 5

6-6. Cell 6: Generation of data for Tableau integration

This cell converts the classification results obtained through the series of cell executions up to this point into an integrated file optimized for visualization in a BI tool (Tableau). Upon executing this cell, "Master Data Export Successful!" is displayed as a message indicating that the corresponding file has been downloaded to the local PC.

6-7. Cell 7: Cross-border Matching Analysis of Corporate Philosophy and IP

This cell extracts corporate philosophies from websites and scores the semantic affinity of the extracted content. As a preliminary task to log in to "Google AI Studio", use the URL "https://aistudio.google.com/app/apikey". After logging in, click "Create a new key". Then, after entering "Gemini API Key" in "Name your key", enter "Default Gemini Project" as

sample information in "Choose an imported project" and click "Create key". Subsequently, copy the "API Key" from "API Key Details".

Next, we will explain the features and screen UI of this cell in Colab (refer to Fig. 11). When selecting "1️⃣ [Classic Method] Upload HTML file and let AI extract automatically". Then, under "Company A (HTML)" and "Company B (HTML)", pre-save the corresponding HTML files from the web and upload them using "Select File" for "Mission", "Select File" for "Value", and "Select File" for "Mind". For reference, the definitions are "Mission" (what is provided to society and customers, reason for existence), "Value" (strengths and commitments in products and services), and "Mind" (action guidelines and attitudes practiced by employees and teams). Next, verify that "Use the initial default prompt" is checked under "Select extraction prompt to use", and click "Run Comparative Analysis" to have an LLM (such as Gemini) automatically extract the single most important sentence corresponding to each definition. After extraction, a compatibility score out of a maximum of 100 points is calculated for each of the "Mission", "Value", and "Mind", and an " Comprehensive Base Score (Ideal)" is also calculated as the average of the three scores. Subsequently, in "Dissatisfied with the extraction results?…", you can enter additional instruction prompts for the initial extraction results, such as "more emotionally clear expressions", and try re-executing by clicking "Apply Additional Instruction and Re-run" using that input prompt. Although options for "Yes" and "No" will be displayed under "Re-matching Verdict after Applying Additional Instructions", clicking "Yes" will display options like "Using previously appended prompt", allowing you to apply the entered additional instruction prompt to future analyses.

On the other hand, when selecting "2️⃣ [Bypass API Limits] Input manually extracted text directly into the form", pre-save the corresponding HTML files from the web and upload them using "Select File" for "Mission", "Select File" for "Value", and "Select File" for "Mind" under "Company A (Direct input)" and "Company B (HTML)". Then, when you click "Run Comparative Analysis", the same output content as when selecting "1️⃣ [Classic Method] Upload HTML file and let AI extract automatically" will be displayed. Incidentally, the output results are the same whether using "Use the initial default prompt" or "Using previously appended prompt".

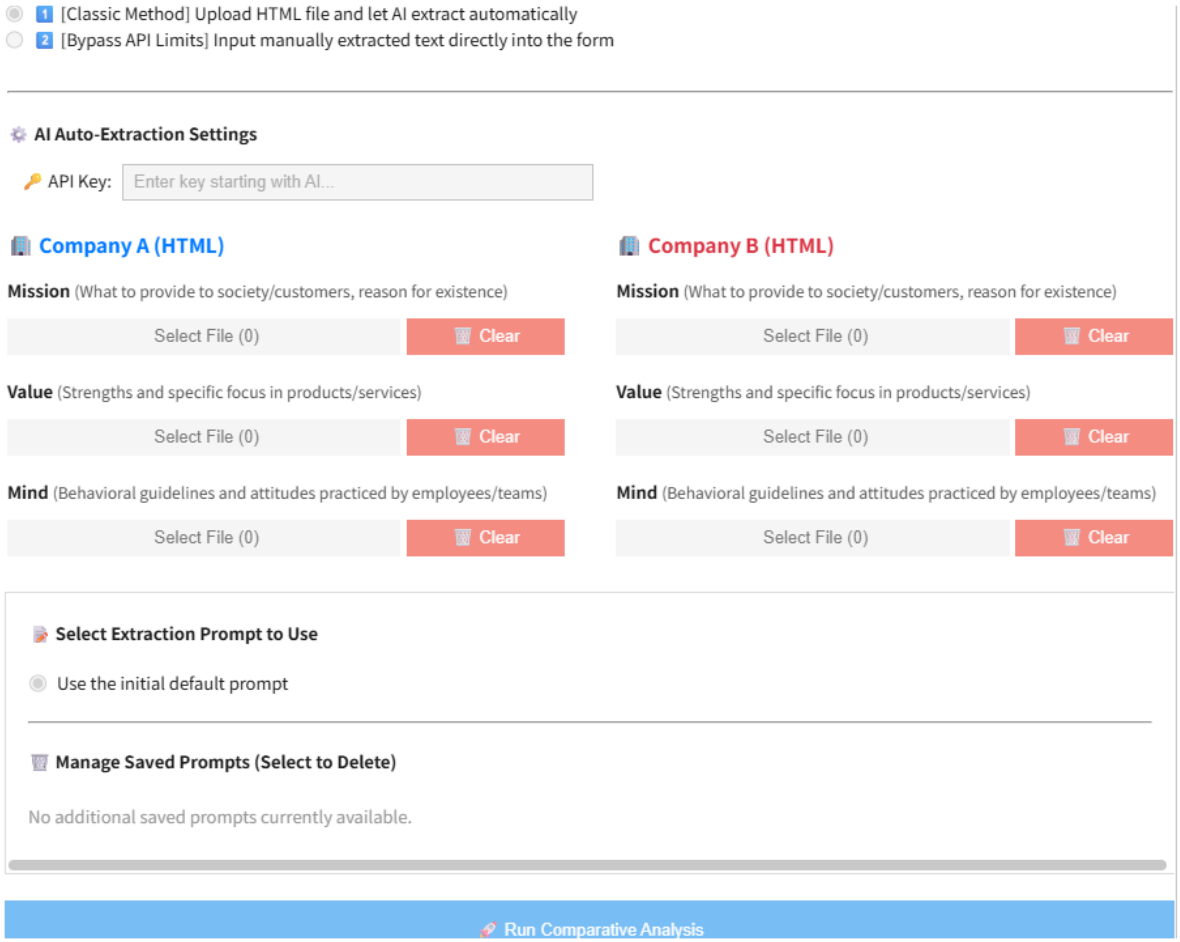


Fig.11 Operation screen for Cell 7

7. Analysis and Construction Methods for Scatter Plots Using Tableau

I will explain the analysis method for visualizing the data output from Colab on Tableau Public using a scatter plot.

7-1. Data Merging in a Local Environment

A process is performed to integrate two data files (e.g., UmaMusume_master_data.xlsx and jal_master_data.xlsx) acquired from Colab to a local environment into a single file. When integrating, first open both files and visually confirm that the header names written in the first row and their arrangement order perfectly match. Next, using one as the base file, transcribe (copy and paste) the data from the other file to be added. At this time, to avoid duplicating headers, extract all cells from the second row to the final row of the file to be added, and paste them into the blank row located directly below the final row of data in the base file.

After merging the data, confirm that both sets of data (e.g., UmaMusume and Jal) are appropriately mixed and stored in the "brand" column within the sheet, and then save it to the local environment by specifying a new file name (e.g., combined_master_data.xlsx).

7-2. Creating a Comment Scatter Plot in Tableau

Sign in to Tableau Public and start a new workbook from "Data Source" under the "Create" menu on your profile page. In "Connect to data" displayed in the center of the screen, upload the merged file (e.g., combined_master_data.xlsx) and check the Scatter plot screen as follows(refer to Fig.12).

The visualization below presents the Japanese analysis output from Colab imported into Tableau. The interface labels are rendered in English via browser auto-translation, Retaining the comments linked to each plot in their original Japanese prevents potential misinterpretation caused by automated translation.Note that the details of the comments associated with each plot are displayed via a hover feature; the procedure for this setup will be described later.

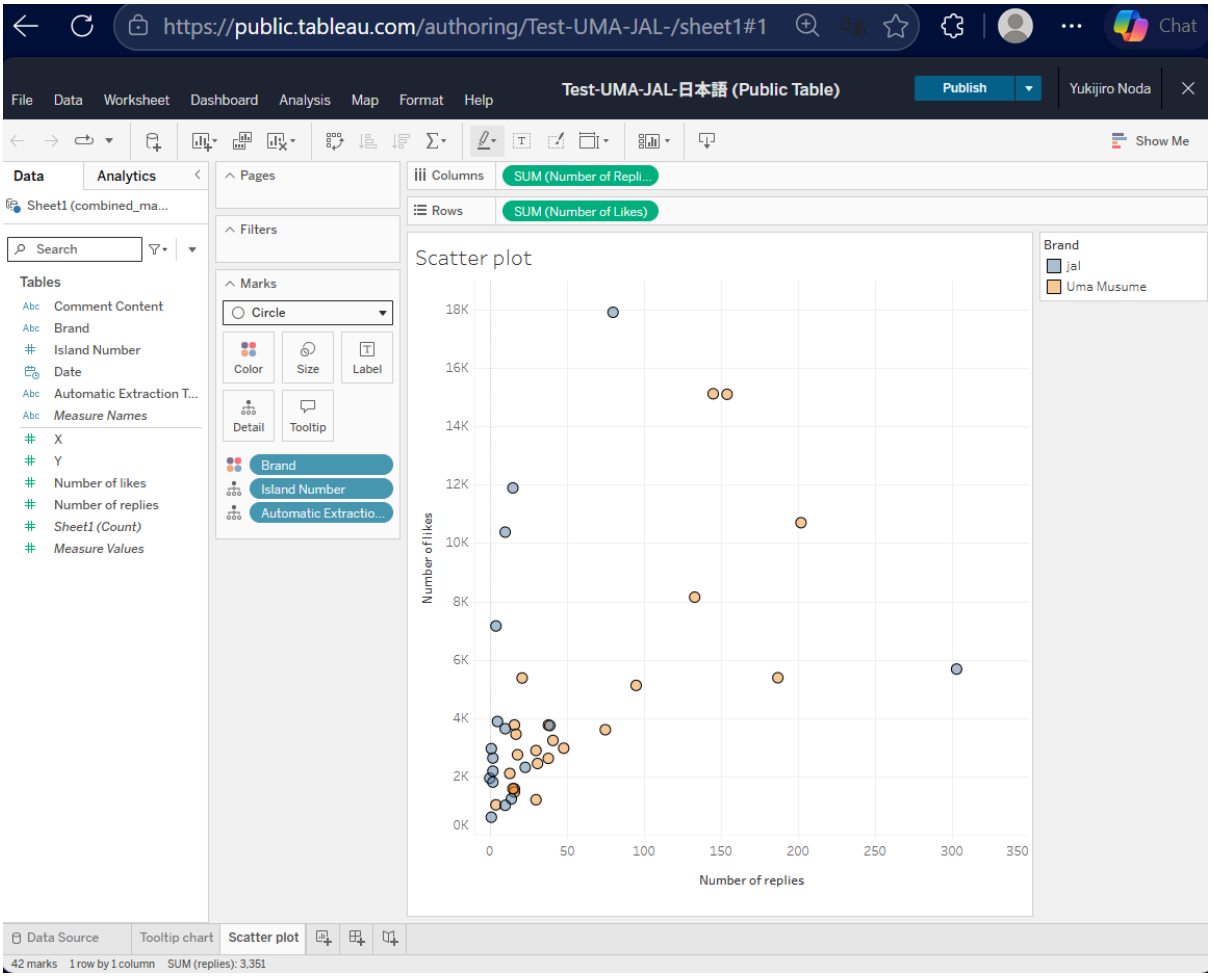


Fig.12 Screen for creating "Scatter plot" in Tableau

Regarding the construction method for Fig. 12, first select "Number of likes" under "Table" on the left side of the screen, and drag and drop it into "Rows" at the top of the screen. Similarly, select "Number of likes" and drag and drop it into "Columns" at the top of the screen. Next, drag and drop "Automatically Extra…" and "Island Number" into "Detail". Drag and drop "Brand" into "Color".Next, customize it so that when you hover the cursor over a plot on the graph, the top 5 comments with the highest "Number of likes" in that cluster are

displayed as a pop-up. Click the "Tooltip" icon, open the text editing dialog box, delete the default text, and write out the layout as follows.

ブランド: <ブランド>
島番号: <島番号>
テーマ: <自動抽出タグ>
<Sheet name="ツールチップ用表" maxwidth="1200" maxheight="400" filter="<All Fields>">

Subsequently, create a new screen as "Tooltip chart" (refer to Fig. 13). Select "Comment Content" under "Tables" on the left side of the screen, and drag and drop it into "Filters" at the top of the screen. Similarly, select "Number of likes" and drag and drop it into "Text". Next, drag and drop "Comment Content" into "Rows". Click "Comment Content" in "Filter" and select "Edit Filter…". Then, in "Top/Bottom", change "Rank" to "Top", change "Count" to "Enter a Value…", change the value to 5, change "Field" to "Number of likes", and set "Aggregation" to "Sum".Click the Tooltip and enter the following text.

コメント本文:　<コメント本文>
いいね数:　<SUM(いいね数)>

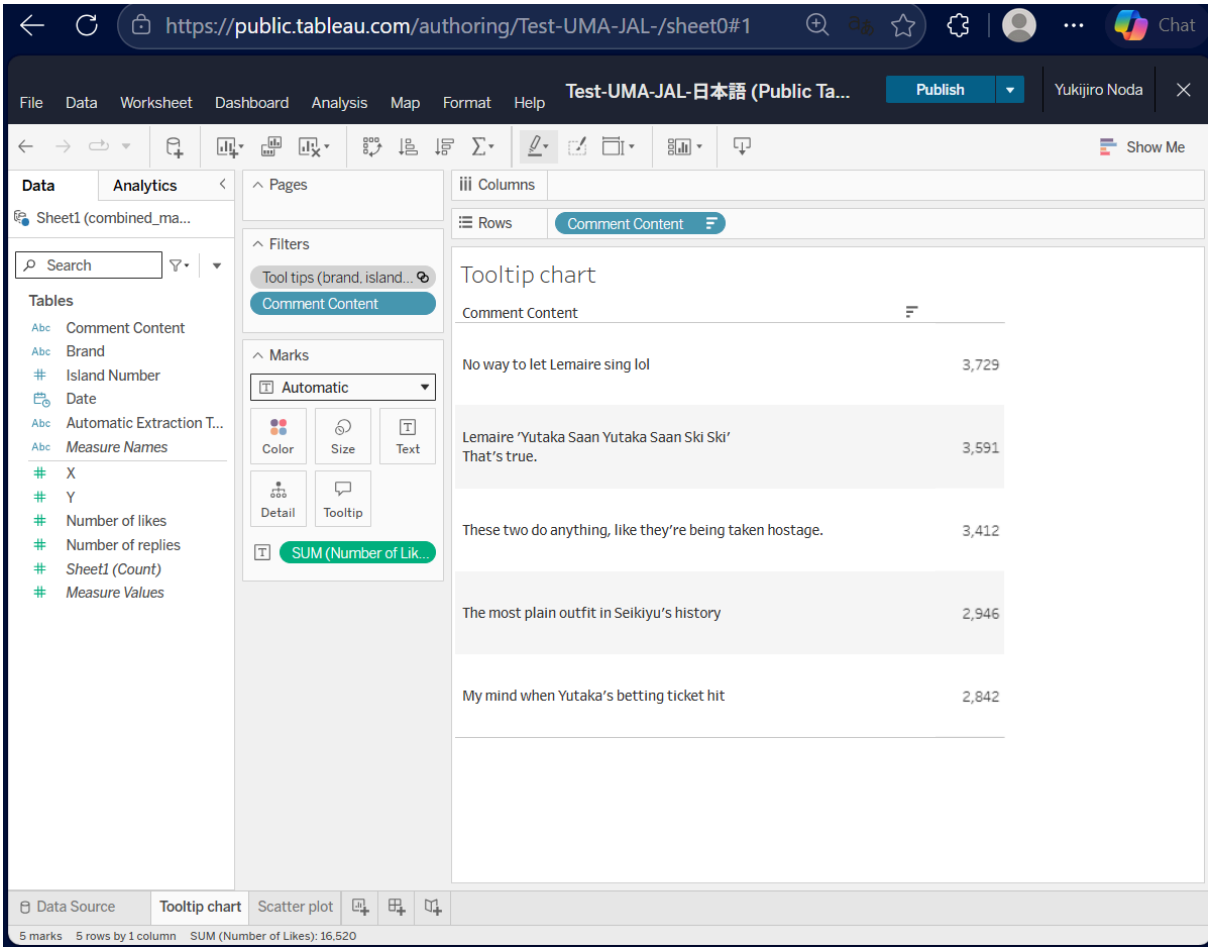


Fig.13 Screen for creating "Tooltip chart" in Tableau